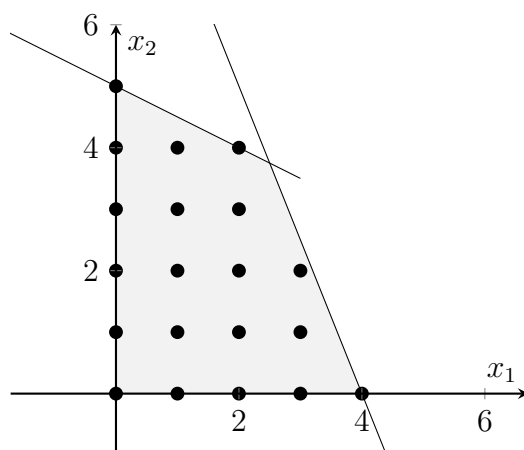


Branch and bound for integer optimization

Michel Bierlaire

Solution of the practice quiz

The feasible set is illustrated in the following figure.



As $(0,0)$ is feasible, we initialize the upper bound to $f^* = 0$. We call P_0 the original problem, and we solve its relaxation:

$$R(P_0) : \min_{x \in \mathbb{N}^2} -13x_1 - 8x_2$$

subject to

$$\begin{aligned} x_1 + 2x_2 &\leq 10, \\ 5x_1 + 2x_2 &\leq 20, \\ x_1, x_2 &\geq 0. \end{aligned}$$

The optimal solution is

$$x_0^* = (2.5, 3.75)$$

with value $f(x_0^*) = -62.5$. We decide to branch on x_1 . We create two subproblems, incorporating a different constraint on x_1 :

- P_1 : $x_1 \leq 2$,
- P_2 : $x_1 \geq 3$.

We now have $\mathcal{P} = \{P_1, P_2\}$

P_1	P_2
$\min -13x_1 - 8x_2$	$\min -13x_1 - 8x_2$
subject to	subject to
$x_1 + 2x_2 \leq 10$	$x_1 + 2x_2 \leq 10$
$5x_1 + 2x_2 \leq 20$	$5x_1 + 2x_2 \leq 20$
$x_1, x_2 \geq 0$	$x_1, x_2 \geq 0$
$x_1, x_2 \in \mathbb{N}$	$x_1, x_2 \in \mathbb{N}$
$x_1 \leq 2$	$x_1 \geq 3$

We consider first P_1 , and solve its relaxation $R(P_1)$. The optimal solution is

$$x_1^* = (2, 4)$$

with value $f(x_1^*) = -58$. It is a feasible solution of the original problem. Therefore, we update the best upper bound to $f^* = -58$. As P_1 has been solved, the set of problems is

$$P_0, P_2.$$

We consider P_2 , and solve its relaxation $R(P_2)$. The optimal solution is

$$x_2^* = (3, 2.5)$$

with value $f(x_2^*) = -59$. This is a lower bound on the optimal value of P_2 . As it is better than the best upper bound ($f^* = -58$), we continue to investigate P_2 by branching. We create two subproblems, incorporating a different constraint on x_2 , which has a non integer value:

- P_{21} : $x_2 \leq 2$,
- P_{22} : $x_2 \geq 3$.

The set of problems is now $\{P_0, P_2, P_{21}, P_{22}\}$

P_{21}	P_{22}
$\min -13x_1 - 8x_2$	$\min -13x_1 - 8x_2$
subject to	subject to
$x_1 + 2x_2 \leq 10$	$x_1 + 2x_2 \leq 10$
$5x_1 + 2x_2 \leq 20$	$5x_1 + 2x_2 \leq 20$
$x_1, x_2 \geq 0$	$x_1, x_2 \geq 0$
$x_1, x_2 \in \mathbb{N}$	$x_1, x_2 \in \mathbb{N}$
$x_1 \geq 3$	$x_1 \geq 3$
$x_2 \leq 2$	$x_2 \geq 3$

We consider P_{21} , and solve its relaxation $R(P_2)$. The optimal solution is

$$x_{21}^* = (3.2, 2)$$

with value $f(x_{21}^*) = -57.6$. As the solution is not integral, this is a lower bound on the optimal solution of P_{21} . But as it is larger than the best upper bound $f^* = -58$, problem P_{21} can be discarded, and the set of problems is

$$\{P_0, P_2, P_{22}\}.$$

Consider problem P_{22} . We observe that it is infeasible, and can be discarded from the set. Therefore, P_2 can also be discarded from the set, and the solution of the original problem is

$$x^* = (2, 4)$$

with value $f(x^*) = -58$.

The process is illustrated as the tree below, where each node corresponds to a subproblem, and the solution of the relaxation and the lower bound are indicated. The nodes in gray correspond to the optimal solution.

